Analysis of Privacy-Focused and Adversary-Favored Operating Systems

Background

With the rapid evolution of cyber-enabled crimes, threat actors are increasingly shifting from conventional operating systems such as Microsoft Windows and macOS to **specialized Linux distributions designed for offensive security, anonymity, and minimal forensic traceability**.

Operating systems such as **Kali Linux, BlackArch, Tails OS**, and similar privacy-centric platforms come preloaded with powerful open-source tools supporting anonymization, reconnaissance, intrusion testing, and covert data transfer. While these tools serve legitimate cybersecurity and research purposes, they are frequently **misused for malicious activities**, creating significant investigative blind spots.

Traditional forensic methodologies remain largely Windows-centric and often fail to provide meaningful visibility into these **non-standard, live-boot, or amnesic operating environments**, where persistent artefacts may be limited or intentionally removed.

Hackathon Challenge

Participants are required to **design and develop a forensic detection and analysis tool or prototype system** capable of performing **operating system–aware forensic inspection** of Linux-based environments commonly used by attackers and privacy-conscious users.

The proposed solution should be capable of operating on:

- A **live Linux system**, or
- A **disk image / filesystem snapshot**

The objective is to assist investigators in **identifying system environments, detecting potentially misused tools, and generating actionable forensic insights** while preserving evidentiary integrity.

Functional Expectations

The developed prototype should be able to:

Operating System Identification

- Detect and identify the underlying Linux distribution
- Recognize offensive-security distributions (e.g., Kali Linux, BlackArch)
- Identify privacy-focused or amnesic systems (e.g., Tails OS, live environments)

Tool & Artefact Detection

Detect installation artefacts, configuration traces, or execution indicators related to:

- **Anonymization & Privacy Tools**
    - Tor services, VPN clients, proxy utilities
- **Reconnaissance & Intrusion Utilities**
    - Network scanners, exploitation frameworks, penetration-testing tools
- **Data Exfiltration & Covert Transfer Mechanisms**
    - Secure transfer or tunneling utilities

Behaviour & Risk Classification

- Classify detected tools and activities into categories such as:
    - High-risk offensive tooling
    - Dual-use cybersecurity utilities
    - Privacy and anonymization infrastructure

Forensic Reporting

Generate a structured forensic report containing:

- Identified operating system characteristics
- Detected tools and associated risk levels
- Supporting artefacts or indicators
- Investigative observations

Design Constraints & Guidance

- The solution **must not rely on decrypting protected content**
- Focus on:
  - Filesystem artefacts
  - Package signatures
  - Metadata
  - Configuration files
  - Process or system traces
- The tool must be **non-destructive and forensic-aware**
- System integrity and lawful investigation principles must be maintained